

INST327 – Assignment 6

Use the iSchool_v2 database for this assignment.

Be sure you complete both questions.

Please consult the rubric and instruction sheets before starting the assignment.

Q.1) Subquery Conversion (50 points)

Rewrite the query shown below to include a subquery. The results for this are listed below the code to be converted.

NOTE: The subquery must NOT be in the SELECT clause; i.e., it should not be included as part of the column list right after the SELECT keyword.

HINT: Consider whether the conditional operation should be “NOT LIKE” or “NOT IN”

```
SELECT MAX(CONCAT(course_code, course_number)) AS max_course_number,  
       course_prereq  
FROM courses  
WHERE course_prereq NOT LIKE "C- (INST201/301)"  
GROUP BY course_prereq  
HAVING max_course_number > "INST500"  
ORDER BY max_course_number;
```

max_course_number	course_prereq
INST610	permission
INST711	INST631 & INST710
INST753	INFM600, INFM603, INST733
INST754	INFM600, INST630, INST733, (INFM603/JOUR652)
INST756	INFM600
INST760	INST630/INFM603
INST775	INST631, INST632, INST717
INST784	INST604

8 rows returned.

Q.2) Subquery / CTE (50 points)

Find each unique combination of building name and earliest course end-time. For buildings where there is more than 1 course start-time, include the number of unique course start-times for that building/end-time combination. Replicate the result set below.

building_name	earliest_course_end_time	unique_course_start_times
Atlantic Building	16:45:00	2
Biology Psychology Building	11:50:00	3
Cambridge Community Center	12:15:00	2
Chemistry-Biochemistry Building	10:45:00	2
Computer Science Instructional Center	11:50:00	2
Engineering Classroom Building (Martin)	09:50:00	3
HJ Patterson Hall	10:15:00	2
Hornbake	14:50:00	2
Hornbake	13:50:00	6
Hornbake	13:45:00	4
Hornbake	12:15:00	4
Hornbake	10:45:00	5
JM Patterson Building	13:50:00	2
John S Toll Physics Building	13:45:00	2
Microbiology Building	12:50:00	3
Online	00:00:00	3
Plant Sciences	10:45:00	2
Skinner Building	13:50:00	2

18 rows returned.

(+5 bonus points): Provide a short explanation of why you used either option, subquery or CTE. Neither answer is more correct, we would just like you to explain your choice. There is no rubric for the bonus question but answers should be more than a simple sentence.